Supplementary Material:
A nuclear-register criterion for radical-pair computation in cryptochrome

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Every number below is computed under the single protocol used in the main text — correlated-pair birth, converged time grid, out-of-sample capacity estimation — with two exceptions that are labelled where they appear and retained deliberately as controls: the first row of Table S2, which is the single-electron reset, and Table S11, a single-realisation clock scan. The sections cover (S1) validation of the Liouville-space solver, (S2) the capacity estimator and its controls, (S3) the injection map and why a single-electron reset fails, (S4) the chemically accessible readout routes, their route-specific memoryless floors, their grid convergence and a re-run of the adverse analyses with the product register, (S5) the coherence fraction at matched hopping rate, (S6) classical baselines, (S7) the turnover clock and the register-reuse criterion, (S8) a semiclassical reference model for the coherence fraction, (S9) which nuclei can actually hold the register, (S10) ensemble averaging over heterogeneous and desynchronised copies, (S11) anisotropic hyperfine coupling and the nuclear spin quantum number, (S12) prediction of the two constants the criterion depends on, and (S13) the spin parameters. Several of Sections S8–S12 *reduce* the claims of the main text; the reductions are what the main text reports. All are reproducible from the open `qbscreen` package.

S1. LIOUVILLE-SPACE SOLVER AND THE COHERENT LIMIT

The propagator is $e^{\mathcal{L}\tau}$ of the Haberkorn master equation assembled in Liouville space with the column-stacking convention $\mathrm{vec}(A\rho B) = (B^{\top}\otimes A)\,\mathrm{vec}(\rho)$; the time-integrated singlet yield follows from a single linear solve,

$$\Phi_S = k_S\,\mathrm{vec}(P_S)^{\dagger}\,(-\mathcal{L})^{-1}\,\mathrm{vec}(\rho_0). \tag{S1}$$

Adding the decoherence term must not corrupt the coherent dynamics. For $1/T_2^e \to 0$ with $k_S = k_T = k$, Eq. (S1) must reproduce the closed-form eigenbasis result $\Phi_S^{\mathrm{coh}} = k\big[\sum_m P_{mm}\rho_{mm}/k + \mathrm{Re}\sum_{m\neq n} P_{mn}\rho_{nm}/(k+i\omega_{mn})\big]$. For a two-electron-plus-nucleus model at $B = 0.9$ mT, $A = 200$ MHz and $k = 1\ \mu\mathrm{s}^{-1}$ the two agree to better than 10^{-6}; this is asserted as a unit test.

S2. CAPACITY ESTIMATOR AND ITS CONTROLS

For a target y_k of the input history the capacity is the out-of-sample squared multiple correlation: a ridge readout ($\lambda = 10^{-6}$) is fitted on the first half of the run and scored on the held-out second half. The totals are $\mathrm{MC} = \sum_{d\geq 0} C[s_{k-d}]$ and $\mathrm{IPC} = \mathrm{MC} + \sum_{\mathrm{nonlin}} C[y]$ over the degree-2 Legendre family (diagonal targets to $d_{\max} = 14$, cross targets for $d_1 < d_2 \leq 6$).

Null floor. Scoring the same feature matrices against targets built from an *independent* input sequence, for which the true capacity is zero, returns $+0.010$ for the estimator used here. The same sum without the $\max(0,\cdot)$ clip returns -1.49, i.e. it is far more biased, in the opposite direction. We therefore retain the clip and quote the floor rather than removing it.

Convergence. The capacity is converged in the length L of the driving sequence (six seeds, engineered point): $\mathrm{IPC} = 5.28\pm0.26$, 5.60 ± 0.20, 5.87 ± 0.19 and 5.84 ± 0.10 for $L = 600$, 1200, 2400 and 4800, i.e. flat beyond $L \approx 2400$. Under the single-electron-reset control the same estimator does *not* converge, rising from 5.9 to 8.1 over the same range; convergence is a property of the drive, not of the estimator.

The Dambre bound is not an independence test. $\mathrm{IPC} \leq n_{\mathrm{obs}}$ holds asymptotically for an orthonormal target family but is violated in finite samples: for $X = s$ ($n_{\mathrm{obs}} = 1$, $T = 1200$) the estimator returns 1.013 ± 0.009 on average and exceeds the bound in 29 of 30 seeds (the stored, regenerable control `s2_dambre_control.json`). Satisfying the inequality therefore does not demonstrate that the readout observables are independent, and we make no such argument anywhere in this work.

The horizon is an integrated delay capacity. The memory horizon quoted in the main text, $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, is the linear capacity summed over delays $d \geq 1$ and multiplied by the cycle time. It is not, in general, the longest delay that can be reconstructed: a reservoir that reproduces only the input one cycle back and one that reproduces only the input ten cycles back have the same $\mathrm{MC} - C_0 = 1$. The two readings agree here only because the delay profile is monotone and short (Table S1): the delay at which the linear capacity falls below 5% of C_0 is 4 turnovers for the CIDNP route and 4 for the kinetic route, against $\mathrm{MC} - C_0$ of 1.82 and 0.93. Every horizon in this paper should be read with that profile in mind; none hides a long tail.

TABLE S1. Linear capacity by delay at the cryptochrome point (survivor register, $q = 0$, six seeds).

delay d (turnovers)	kinetics (5 ch)	+CIDNP (8 ch)
0	1.000	1.000
1	0.662	0.846
2	0.184	0.466
3	0.051	0.177
4	0.020	0.036
5	0.005	0.033
6	0.001	0.039

S3. THE INJECTION MAP

A protocol that resets only the input electron to a product state while retaining the reduced state of the rest is not a chemical process. For a separated pair ($J = 0$) it destroys the electron–electron correlation on every cycle, and since $\langle P_S\rangle = \frac{1}{4} - \langle \boldsymbol{S}_1 \cdot \boldsymbol{S}_2\rangle$ with $\langle \boldsymbol{S}_2\rangle = 0$, the singlet probability is pinned at exactly 1/4 independently of the input. We replace it by correlated-pair birth: the nuclei are retained and the electron pair is recreated as $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$, or as a coherent S–T_0 superposition. Table S2 gives the consequence at the cryptochrome point ($B = 50$ μT, $J = 0$, $\tau = 1$ μs, six seeds, kinetic-only readout).

TABLE S2. The injection map and the capacity it permits.

injection	max. flux	IPC
single-electron reset	$1.2e-13$	0.000 ± 0.000
correlated-pair birth, S/T	2.4×10^{-1}	2.543 ± 0.120
correlated-pair birth, S–T_0	3.7×10^{-1}	3.218 ± 0.147

Under the single-electron reset the spread of the observable is at the level of machine precision: the readout is a constant, and the vanishing capacity is an identity of the map rather than a statement about spin chemistry.

What carries the input. The injection map above encodes the input in the singlet character s of the pair at birth. Table S3 drives the same reservoir, with the product register carried, through carriers that chemistry or physics could modulate: the field magnitude ($B = 25 + 50u$ μT, singlet birth), the singlet recombination rate ($k_S = 0.5 + u$ μs^{-1}, singlet birth) and a small-signal $s = \frac{1}{2} + \delta(u - \frac{1}{2})$. Inputs are quantised to 32 levels so that one propagator per level is built once; every encoding is scored against its own floor, the register wiped every cycle (six seeds).

TABLE S3. Capacity by input carrier at the cryptochrome point (product register, six seeds).

carrier	5 ch	floor	8 ch	floor	excess
s, full range	2.771	1.019	4.962	2.005	+2.957
s, $\delta = 0.1$	1.309	1.019	2.810	1.060	+1.750
s, $\delta = 0.01$	1.011	1.019	1.781	1.019	+0.762
s, $\delta = 0.001$	0.681	0.686	0.703	0.695	+0.008
B (25–75 μT)	2.984	1.996	5.144	2.004	+3.140
k_S (0.5–1.5 μs^{-1})	2.051	2.006	4.425	2.006	+2.419

The field is the strongest carrier of the three and the rate carrier matches the s encoding on the CIDNP route; the kinetic route, which reads the time course of one yield, carries little under a rate input because the rate reshapes the time course of every cycle alike. The small-signal rows show the other dependence: the excess falls with the input amplitude and is gone at $\delta = 10^{-3}$, where the channel variations drop below what the ridge of Sec. S2 resolves. The floors themselves fall there for the same reason. The criterion is therefore not tied to an s that nothing modulates, but it is tied to an input of order the full dynamic range.

S4. CHEMICALLY ACCESSIBLE READOUT ROUTES

Recombination is switched on ($k_S = 1$, $k_T = 0.2$ μs^{-1}) so that the observable is the yield actually formed, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\,\mathrm{d}t$, sampled at five laboratory times. CIDNP is the product-weighted nuclear polarisation, $\langle I_z\rangle_{\mathrm{prod}} = \int k_S \mathrm{Tr}[I_z P_S \rho]\,\mathrm{d}t \,/\, \int k_S \mathrm{Tr}[P_S \rho]\,\mathrm{d}t$: the polarisation carried away by the recombined molecule, not that of the surviving pair.

TABLE S4. Capacity recovered by chemically accessible observables (six seeds).

readout route	channels	IPC
end-point singlet yield	1	1.018
both exit channels, end point	2	1.283
three micro-environments (different τ)	3	2.722
time-resolved kinetics, one product pool	5	1.971
both exit channels, time-resolved	10	1.974
kinetics + accumulated downstream pool	10	2.817
kinetics + CIDNP on the product	8	4.421

All seven routes defined by the model are listed. The last two rows are separate constructions, not cumulative additions: the CIDNP route is five kinetic channels plus three polarisation channels, hence eight, not eighteen. The omitted route is informative: doubling the time-resolved channel count from five to ten by adding the second exit channel changes the capacity by 0.003, so the readout saturates in channel number.

Route-specific memoryless floors. The floor is not common to these routes and is measured by wiping the register every cycle ($q = 1$), which leaves only what each route computes from the present input (six seeds): 1.019 ± 0.025 for the one-, two-, three-, five- and ten-channel yield routes (including the heterogeneous-lifetime route, whose floor is measured by the same $q = 1$ construction), 2.005 ± 0.017 for the eight-channel CIDNP route, and 1.960 ± 0.070 for the accumulated pool. The CIDNP excess is therefore $4.421 - 2.005 = 2.42$, not the factor of 4.4 that a universal floor of unity would imply. The accumulated-pool route retains MC = 1.937 at $q = 1$ because a leaky integrator is itself a classical delay line; its capacity is not evidence of nuclear-spin memory and we exclude it from the claims in the main text.

The register is the product. The cycle above hands the next turnover the renormalised state of the pair that did *not* recombine. Rebuilding it so that the register is the nuclear state carried away by the recombined molecule, $\sigma_{\mathrm{nuc}} \propto \int [k_S \mathrm{Tr}_e(P_S \rho P_S) + k_T \mathrm{Tr}_e(P_T \rho P_T)] dt$, raises every capacity: kinetics 1.971 → 2.771, both-exit time-resolved 1.974 → 2.779, CIDNP 4.421 → 4.962, accumulated pool 2.817 → 4.063 (six seeds). The reported numbers are thus conservative with respect to this choice.

Do the adverse results survive the product register? They do. The sensitivity analyses of Secs. S9–S11 are quoted for the survivor register, so that they are directly comparable with the published pipeline; each has also been re-run with the product register.

TABLE S5. The reductions are not artifacts of the survivor register.

quantity	survivor	product
proton register, excess retained	32%	29%
proton register, kinetic excess	+0.000	+0.000
kinetic excess at 2 cycles of jitter	+0.007	−0.021
CIDNP excess at 2 cycles of jitter	+1.019	+1.156
anisotropy $\eta = 0.5$, fraction retained	57%	35%
anisotropy $\eta = 1$ (Ising), excess	0.000	0.009

Every reduction survives, and none of the paper's conclusions depends on the choice. The one visible difference is that the kinetic readout degrades more gracefully under moderate jitter with the product register (excess +0.353 at half a cycle, against +0.000), though it still reaches zero by two cycles. Anisotropy is more punishing with the product register (35% retained against 57%), and the Ising point still gives an excess of 0.009, i.e. zero within the estimator's floor.

Time-grid convergence. A 40 MHz hyperfine coupling has a 25 ns period, so a coarse grid under-samples the dynamics and inflates the time-resolved capacities. Table S6 shows convergence from $n_t = 48$ onward; the main text uses $n_t = 96$ ($\mathrm{d}t \approx 10$ ns).

TABLE S6. Convergence in the number of time steps per cycle (six seeds, $L = 700$, the same protocol as Table S4).

n_t	dt (ns)	kinetics IPC	+CIDNP IPC
24	41.7	2.244	4.456
48	20.8	1.996	4.423
96	10.4	1.971	4.421
192	5.2	1.972	4.427

Why CIDNP is not an optional extra channel. The reduced state of either electron of a newly born pair is maximally mixed for every input, because both $|S\rangle\langle S|$ and $P_T/3$ carry zero one-electron polarisation; the input resides entirely in the two-electron correlation. With $J = 0$ the radicals do not interact, so nothing transfers that correlation to a nucleus. Consistently, with recombination switched off the cryptochrome-point capacity is 1.01, the memoryless floor. Spin-selective recombination is what converts the singlet–triplet correlation into a population difference and so writes the input into the nuclear register: in a separated pair it is the write mechanism, not merely the read mechanism.

The cycle closed over the molecule-number budget. Both registers above are conditional. The survivor is the pair that had *not* recombined by τ, renormalised; the product is the nuclear state of the molecules that *had*, renormalised. At $k_S = 1$, $k_T = 0.2\ \mu\mathrm{s}^{-1}$ and $\tau = 1\ \mu$s the fraction still unreacted at τ is 0.66, so neither is the register of the whole population. The closed carrier sums the two branches with their molecule-number weights before renormalising: every molecule hands its register on, whichever branch it took, the late recombination of the unreacted branch being taken to leave its nuclear state as it stands at τ. A regeneration probability below one, some molecules being replaced by fresh ones, is the register depolarisation q already scanned. Table S7 compares the three carriers for the full and the proton-only register (six seeds; floors 1.019 and 2.005 throughout).

TABLE S7. Capacity by register carrier at the cryptochrome point.

register	carrier	5 ch	8 ch	excess
all three	survivor	1.971	4.421	+2.416
all three	product	2.771	4.962	+2.957
all three	closed	2.336	4.695	+2.690
proton only	survivor	1.019	2.784	+0.779
proton only	product	1.019	2.866	+0.861
proton only	closed	1.019	2.560	+0.555

The closed register lies between the two conditional ones for the full register (excess 2.69 against 2.42 and 2.96) and below both for the proton-only register (0.55 against 0.78 and 0.86): the two branches carry polarisations of opposite sign, as spin sorting requires, and

summing them cancels part of what either branch alone would hand on. The proton-register criterion is therefore recomputed on the closed carrier in Sec. S9.

S5. COHERENCE FRACTION AT MATCHED HOPPING RATE

An all-spin pure-dephasing rate γ is added in the S_z product basis. Taking $\gamma \to \infty$ at fixed τ measures Zeno freezing rather than a classical limit, because the golden-rule transfer rate scales as $|H_{ij}|^2/\gamma$; we therefore hold rate $\times\, \tau$ fixed by scaling $\tau \propto \gamma$, whereupon the capacity saturates.

TABLE S8. Incoherent limit at matched hopping rate (six seeds).

γ (MHz)	τ (ns)	IPC
0 (coherent)	50	5.421
10^3	10^3	0.999
5×10^3	5×10^3	0.998
2×10^4	2×10^4	0.998

The saturation value 1.00 is the memoryless floor, so the coherence fraction is $(5.421 - 0.999)/5.421 = 0.82$.

The added γ above acts on all five spins, which raises the objection that the collapse merely records erasure of the nuclear register rather than loss of electronic coherence. It does not. Restricting the added dephasing to the two electrons, leaving the nuclei entirely untouched, and holding the same matched τ gives IPC $= 0.9949 \pm 0.0222$, 0.9891 ± 0.0223 and 0.9891 ± 0.0224 at $\gamma = 10^3$, 5×10^3 and 2×10^4 MHz (six seeds), i.e. a coherence fraction of 0.817 against 0.816 for the all-spin channel. The quantity is a statement about the electrons.

It remains, however, an *operational* fraction: it is defined by one particular incoherent limit in which τ is stretched by up to a factor of 400, not by comparison against a physically motivated classical model. The appropriate reference would be a semiclassical radical-pair treatment with classical hyperfine vectors and rate equations [Timmel *et al.*, Mol. Phys. **95**, 71 (1998)]. Section S8 supplies exactly that reference and finds the classical model at its own memoryless floor: it holds no memory the quantum system lacks. Its absolute capacity is trajectory-sampling limited, so no coherence fraction is derived from it, and we do not present 0.82 as a measure of "how quantum" the reservoir is.

S6. CLASSICAL BASELINES

Tanh echo-state networks (spectral radius 0.9, input scale 0.6) are driven by the same input and scored with the same estimator (eight seeds, $L = 1200$).

TABLE S9. The quantum reservoir against matched classical reservoirs.

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.593 ± 0.191
ESN, 5 nodes (5 linear)	4.552 ± 0.325
ESN, 5 nodes (12: 5 linear + 7 quadratic)	9.414 ± 0.860
ESN, 12 nodes (12 linear)	10.366 ± 0.590

Matched by physical unit the quantum system is ahead; matched by readout dimension it is behind. We report both and claim no advantage. Which accounting is fair depends on a cost model for non-commuting measurements against multiplications that we do not have.

The cryptochrome-point comparison and the regulariser. The channel-matched comparison at the cryptochrome point (main text, Table III) is sensitive to the ridge in a way the engineered-point comparison is not: the chemical channels have amplitudes of order 10^{-2}, the ESN states of order one, and a single absolute $\lambda = 10^{-6}$ penalises only the former (Sec. S7). Table S10 repeats the comparison with the features standardised on the training half and λ selected per target by five-fold cross-validation on that half, from the grid 10^{-12} to 1, for both systems and over the same twelve input seeds.

TABLE S10. Channel-matched comparison at the cryptochrome point with the ridge fixed (six seeds) and with the ridge chosen by cross-validation (twelve paired seeds). The last two rows are paired differences, ESN minus quantum, with their SEM.

system (readout features)	$\lambda = 10^{-6}$	λ by CV
quantum, kinetics (5 ch)	2.02 ± 0.13	4.64 ± 0.29
ESN, 5 nodes	4.56 ± 0.30	4.55 ± 0.30
quantum, +CIDNP (8 ch)	4.54 ± 0.25	6.90 ± 0.47
ESN, 8 nodes	6.96 ± 0.62	6.95 ± 0.61
ESN − quantum, 5 ch	$+2.53 \pm 0.07$	-0.10 ± 0.10
ESN − quantum, 8 ch	$+2.42 \pm 0.17$	$+0.06 \pm 0.16$

With the regulariser chosen the same way for both, the classical margin is gone at both channel counts. The fixed-λ figures are kept in the main text because every other capacity in this paper is computed at that λ; the conclusion that there is no quantum advantage holds under either, and the conclusion that the classical system is better holds under neither.

S7. TURNOVER CLOCK AND THE REGISTER-REUSE CRITERION

The memory horizon is $(\mathrm{MC} - C_0)$ times the cycle time, the zero-delay term $C_0 \simeq 1$ contributing no memory; the raw $\mathrm{MC} \times T$ used in Table S11 below is retained there as the single-realisation control. Because the nuclei leave in

the diamagnetic product the cycle time is the turnover interval T_{d} and not the radical-pair lifetime. We insert a pause between turnovers and relax the register with probability $q = 1 - \exp(-T_{\mathrm{d}}/T_{1n})$, taking $T_{1n} = 1$ s. That value is the high-field literature figure, used here so that the scan isolates the clock; Sec. S12 shows it is a high-field number and that the correct low-field value for a protein-bound proton is some four hundred times shorter, which is what ultimately binds the criterion.

TABLE S11. The horizon follows the turnover interval.

T_{d}	q	MC (5 ch)	MC (8 ch)	horizon (8 ch)
1 μs	10^{-6}	1.887	2.756	5.5 μs
1 ms	10^{-3}	1.887	2.755	2.8 ms
10 ms	10^{-2}	1.889	2.747	27 ms
100 ms	9.5×10^{-2}	1.899	2.606	261 ms

Table S11 is a *single*-input-realisation scan; four significant figures are not supportable on that sample, and its 0.2% comes from the five-channel column whereas the horizon is an eight-channel quantity. Repeating the scan with twelve realisations gives MC (8 ch) = 2.919 ± 0.205, 2.916 ± 0.204, 2.894 ± 0.196 and 2.679 ± 0.124 for the same four intervals. Since the same seeds are used at every T_{d}, the change is assessed on paired differences: 0.239 ± 0.028 (SEM, $n = 12$, $t = 8.46$), i.e. 8.2% across five decades. That is a real decrease, resolved above both the seed noise and the grid error, and concentrated at the longest pause: the first three decades cost under one per cent and the last decade the rest. A 10^5-fold change in the clock moves the capacity by under a tenth, so the horizon tracks the clock. Table S12 isolates the requirement that actually binds: how much depolarisation of the register between turnovers the memory tolerates.

The linear memory capacity, which is the quantity these claims concern, also converges in the time grid, as the IPC does: MC (8 ch) = 2.790, 2.820, 2.821 for $n_t = 48$, 96, 192 (six seeds), a 0.03% change across the last refinement.

TABLE S12. The criterion: memory against register depolarisation q.

q	T_{d}/T_{1n}	MC (5 ch)	MC (8 ch)
0.00	0	1.941	2.846
0.30	0.36	1.926	2.330
0.50	0.69	1.889	2.088
0.70	1.20	1.707	1.997
0.90	2.30	1.163	1.933
1.00	∞	1.002	1.002

Memory degrades steadily as the turnover interval approaches the nuclear relaxation time—by $q = 0.5$ more than a quarter of the eight-channel capacity is already gone—and collapses to the memoryless floor MC = 1.00 when every turnover starts from a fresh molecule.

Which nuclear relaxation process enforces the criterion. The register propagated between turnovers is a full nuclear density matrix, so memory might in principle require nuclear coherences as well as populations to survive the pause; the binding constant would then be $\min(T_{1n}, T_{2n})$, which at NMR fields is much the more restrictive. Replacing the depolarising channel by a pure-dephasing channel on each nucleus—which destroys nuclear coherence while preserving $\langle I_z \rangle$ exactly—settles it (four seeds):

TABLE S13. The register is a population register, not a coherence register.

q	MC (8 ch), depolarising	MC (8 ch), dephasing
0.0	2.846	2.846
0.3	2.330	2.749
0.5	2.088	2.722
0.7	1.997	2.715
0.9	1.933	2.716
1.0	1.002	2.716

Complete nuclear dephasing every cycle costs less than 1% of the capacity. The memory resides in nuclear z-populations—precisely what the CIDNP readout reports—so the constant that bounds the horizon is T_{1n}.

This is a statement about what the reservoir uses, not a rescue. At the geomagnetic field the register is in extreme narrowing, where $J(0)$, $J(\omega)$ and $J(2\omega)$ coincide and therefore T_{1n} and T_{2n} agree to one part in 10^7 (Sec. S12); the population/coherence distinction changes nothing there, and would matter only in an NMR magnet.

Regularisation and counting statistics. The ridge parameter $\lambda = 10^{-6}$ on unstandardised features is load-bearing, because the time-resolved channels are nested partial integrals of a single signal and the Gram matrix is correspondingly ill-conditioned ($\kappa \approx 10^{10}$). For the kinetic route the recovered IPC runs 4.50, 3.61, 1.93, 1.01, 1.00 at $\lambda = 10^{-12}$, 10^{-9}, 10^{-6}, 10^{-3}, 10^{-1}; standardising the features gives 4.50, 4.50, 3.69, 2.22, 1.39. The reported capacities are therefore conservative with respect to this choice, not inflated by it. The physically meaningful regulariser is the noise on a real product pool, counted as a pool is counted. A yield is Binomial$(N, y)/N$ for N molecules, the five cumulative yields being nested counts of one pool, and a product polarisation is the mean of Ny_S values of ± 1 drawn with probability $(1 \pm P)/2$, so its variance is $(1 - P^2)/Ny_S$ and does not vanish with P. Table S14 resamples the eight channels this way for the full register and for the proton-only product register, with each register's floor ($q = 1$) counted under the same noise, and sets the result beside the relative-Gaussian model ($N^{-1/2}$ times the channel's mean magnitude) that was the natural first guess.

TABLE S14. CIDNP-route excess over the counted floor against the number of molecules read as one pool (six seeds); the last column is the relative-Gaussian model $N^{-1/2}$ on the full register. Noiseless excess: 2.42 (full register), 0.86 (proton-only product register).

N	full, counted	proton, counted	full, Gaussian
10^5	0.121	−0.010	1.634
10^6	0.493	−0.005	1.800
10^7	0.956	0.104	2.224
10^8	1.933	0.542	2.395
10^9	2.352	0.814	2.412
10^{10}	2.411	0.855	2.416
10^{12}	2.416	0.861	2.416

The two noise models part company below 10^9: at 10^8 the counted excess is 80% of the noiseless value for the full register and 63% for the proton register, where the Gaussian model still shows 99%; at 10^6 the counted excess is gone and the Gaussian model retains 74%. The readout is counting-noise limited below about 10^9 molecules per pool.

S8. A SEMICLASSICAL REFERENCE FOR THE COHERENCE FRACTION

The coherence fraction of Sec. S5 is defined by an operational dephasing limit, not by comparison against a physically motivated classical model, and we said there that it should not be read as a quantum/classical decomposition without one. This section supplies the reference.

We keep the Hamiltonian, couplings, drive, readout and estimator identical and change one thing: every spin becomes a classical vector of length $\sqrt{S(S+1)}$ precessing in the field of the others, $\dot{\boldsymbol{S}}_i = \boldsymbol{S}_i \times \partial H/\partial \boldsymbol{S}_i$. This is the Schulten–Wolynes/Timmel semiclassical treatment carried to its natural conclusion—classical nuclei *with* back-action, so that the register can still carry memory between turnovers. A static-nucleus semiclassical model would have no memory by construction and would be a straw man. The correspondences are exact in the mean: $\langle P_S\rangle = \frac{1}{4} - \boldsymbol{S}_1\cdot\boldsymbol{S}_2$, singlet birth is an antiparallel pair ($P_S = 1$), triplet birth is a fixed angle $\cos\theta = 1/3$ ($P_S = 0$), and electronic T_2^e becomes a white-noise z-field. Each of $M = 512$ trajectories is one molecule carrying its own register, so the trajectory average is simultaneously the classical reference and a pooled ensemble of independent registers.

Two checks: the sampled $\langle P_S\rangle$ reproduces the input s to four decimals across $s \in [0,1]$, and $|\boldsymbol{S}|$ is conserved to 5×10^{-7} over a cycle. The capacity shows no systematic drift with the integration step (0.78, 0.88, 0.73 for 3, 6 and 12 substeps per output point); the scatter is Monte-Carlo noise in the pooled observable, not integration error.

At the engineered point with the same twelve observables (four seeds, $L = 700$, $M = 512$):

$$\mathrm{IPC}_{\mathrm{classical}} = 0.764 \pm 0.015, \qquad \mathrm{IPC}_{\mathrm{quantum}} = 5.421.$$

The classical value sits below the nominal memoryless floor of unity: with 512 trajectories the pooled observable cannot reconstruct even the current input exactly, so the absolute number is limited by trajectory sampling rather than by the physics of the model. The comparison that is meaningful at finite M is therefore against the classical model's *own* floor, obtained by randomising its register every cycle and sampled with the same M and the same seeds:

$$\begin{aligned}\mathrm{IPC}_{\mathrm{classical}} &= 0.764 \pm 0.015,\\ \mathrm{IPC}^{q=1}_{\mathrm{classical}} &= 0.768 \pm 0.016,\\ \text{paired excess} &= -0.004 \pm 0.016.\end{aligned}$$

The excess is zero within the noise. The classical reservoir retains *no* memory of previous turnovers, and its linear and total capacities coincide to three decimals, so it has no nonlinear capacity either. Classical nuclei with full back-action, given every opportunity to store information between cycles, store none. Table S15 records the live and floor values as the trajectory number is raised; the excess is what should converge, and the absolute values are expected to rise towards unity as the sampling noise in the current input falls.

TABLE S15. Trajectory-number scan of the semiclassical reference (four seeds, live and floor paired per seed).

M	IPC, live	IPC, floor ($q = 1$)	paired excess
512	0.764 ± 0.017	0.768 ± 0.019	-0.004 ± 0.016
2048	0.941 ± 0.011	0.935 ± 0.010	$+0.006 \pm 0.016$

What this reference supports is the statement that the classical model holds no memory the quantum one does not. What it does not support, at the trajectory numbers we can afford, is a converged coherence fraction: the ratio $(5.421-0.764)/5.421 = 0.859$ would move as the classical absolute value converges, and we therefore do not quote it as a result. The 0.82 of the main text is the operational dephasing limit and is reported as such.

S9. WHICH NUCLEI CAN HOLD THE REGISTER

The main text applies a single relaxation probability q to the whole nuclear register, with $T_{1n} = 1$ s. The relaxation literature does not support treating the three nuclei alike, and the distinction matters more than any other in this work.

What is measured. For protein-bound ^{1}H the assumption is sound: non-selective amide T_1 in ubiquitin is 1.15 s at 600 MHz [Tjandra *et al.*, J. Am. Chem. Soc.

117, 12562 (1995)], and 3.0–18.7 s in a deuterated domain [Ulmer *et al.*, J. Magn. Reson. **166**, 190 (2004)]. In vivo ^{31}P in human brain and muscle spans 1.1–6.7 s [Bogner *et al.*, Magn. Reson. Med. **62**, 574 (2009); Ren *et al.*, NMR Biomed. **28**, 1455 (2015)]. For ^{14}N it is not sound. Nitrogen-14 is spin-1 and relaxes quadrupolarly; in immobilised proteins ^{14}N acts as a fast-relaxing relaxation sink [Sunde and Halle, J. Magn. Reson. **203**, 257 (2010)], and rotationally immobilised protein protons already show $1/T_1 > 2000$ s^{-1} at low field [Goddard, Korb and Bryant, J. Magn. Reson. **199**, 68 (2009)]. Using the measured quadrupole tensors of the flavin semiquinone—$e^2qQ/h \approx 3.6$ MHz for N5 and 4.8 MHz for N10 [Martínez *et al.*, Magn. Reson. **6**, 183 (2025)]—with a rotational correlation time of ~ 20 ns for a cryptochrome-sized protein, and noting that at geomagnetic field $\omega\tau_c \ll 1$ places the system firmly in extreme narrowing where quadrupolar relaxation is *maximal*, gives $T_1(^{14}\mathrm{N}) \approx 0.14$–$0.26$ μs. Even the most favourable measured biological ^{14}N coupling reaches only ~ 1.3 μs.

Two of the three nuclei in our model are ^{14}N5 and ^{14}N10 on FAD. For them, $q = 1 - \exp(-T_\mathrm{d}/T_{1n}) \to 1$ for any turnover interval beyond a microsecond: they are wiped every cycle, whatever the turnover rate.

We note for completeness that the objection usually raised—paramagnetic relaxation from dissolved O_2 and cytoplasmic metal ions—is *not* what binds here. Measured oxygen relaxivity is 0.14 s^{-1}mM^{-1} on water protons at ≥ 3 T [Vatnehol *et al.*, MAGMA **33**, 447 (2020)] and 0.03–0.5 s^{-1}mM^{-1} on protein-bound protons depending on burial [Teng and Bryant, Biophys. J. **86**, 1713 (2004)]; at a tissue [O_2] of ~ 30 μM, and with cytosolic labile Fe at 0.2–7 μM, the total paramagnetic contribution is $\lesssim 0.1$ s^{-1}, a ceiling of tens of seconds. The bound on the proton register is set by slow tumbling, not by paramagnets.

What survives. We therefore relax each nucleus independently, depolarising nucleus j by $\mathrm{Tr}_j(\rho) \otimes \mathbb{1}_j/2$, and score every scenario against the all-wiped floor of the same readout (six seeds).

TABLE S16. Capacity by which nuclei retain their polarisation, as excess over the all-wiped floor.

register retained	kinetics	excess	CIDNP	excess
all three (main text)	1.971	+0.952	4.421	+2.416
^{1}H only (literature)	1.019	+0.000	2.784	**+0.779**
^{14}N only	1.036	+0.017	3.061	+1.056
none (floor)	1.019	0	2.005	0

The restriction is severe, and we state it as such. A physically defensible register—the proton alone—retains +0.779 of excess capacity, 32% of the value obtained when all three nuclei are assumed to persist. The time-resolved kinetic readout is destroyed outright: it needs the whole register and returns exactly its floor when either subset is wiped. (That is a statement about the survivor register: with the product register the kinetic excess still falls to zero when the ^{14}N are wiped, but wiping the proton alone leaves +0.48.) **The reservoir therefore survives on the proton register, at roughly a third of the capacity, and only through the CIDNP channel.** The proton T_{1n} of ~ 1 s that makes this seem comfortable is, however, a *high-field* measurement; Sec. S12 shows that at the geomagnetic field the same nucleus relaxes some four hundred times faster, which tightens the criterion further. What shrinks here is how much the reservoir can compute and how it must be read; what Sec. S12 shrinks is the window in which it can be read at all.

This also sharpens the experimental target: the criterion should be tested on a radical pair whose register is protons, and the relevant relaxation time is the proton T_1 of the diamagnetic product, not a generic "nuclear T_1".

The register-reuse curve of the register the biology keeps. Every ceiling and boundary in Sec. S12 feeds the proton T_1 into the curve of Table S12, which relaxes all three nuclei together and carries the survivor. The two restrictions established above—^{14}N wiped every cycle, product carried—and the proton's partial relaxation must be imposed together for the boundary to be about the register it names. Table S17 does so: the two ^{14}N are depolarised with probability one, the product register is handed on, and the proton is depolarised with probability q_H on the same grid as the refined scan (six seeds).

TABLE S17. Memory against proton depolarisation q_H for the proton-only product register (^{14}N wiped every cycle).

q_H	MC (8 ch)	IPC (8 ch)
0.00	1.850 ± 0.063	2.866
0.30	1.886 ± 0.040	2.900
0.50	1.890 ± 0.034	2.904
0.70	1.823 ± 0.021	2.836
0.80	1.668 ± 0.020	2.678
0.90	1.298 ± 0.020	2.301
0.94	1.120 ± 0.020	2.122
0.98	1.010 ± 0.011	2.012
1.00	1.003 ± 0.004	2.005

The curve is flatter than the full register's at small q_H—the proton alone holds less, but what it holds it keeps to $q_\mathrm{H} \simeq 0.5$—and its floor, 1.003, is used as C_0. Running the ceiling and the bisected boundary of Sec. S12 on this curve gives, for the Trp Hβ vector, a ceiling of 2.6 ms on the intramolecular term and 1.7 ms with the bath (against 5.4 and 3.5 ms for the full register) and boundaries of 4.0 and 2.6 ns (against 8.3 and 5.4 ns); the required speed-up over the protein's own tumbling is 3.8× and 5.9×. For the flavin 8α-CH_3 the ceiling is 5.4 ms without the bath and 3.0 ms with it, so the methyl escape of Sec. S12 closes on this register without appeal to the bath; its boundaries are 8.2 and 4.6 ns. These

are the numbers the criterion should be read with; the full-register values bound what the spin dynamics would permit if every nucleus could be kept.

Closing the molecule budget. The curve above hands on the product register of the molecules that recombined by τ, which is a third of them (Sec. S4). Two ways of closing the cycle over the other two thirds bracket what the biology could do. If only the recombined branch is regenerated and the rest are replaced by fresh molecules every cycle, the proton is handed on by a fraction $f = 0.34$ of the pool and the effective depolarisation is $q_{\mathrm{H}} = 1 - f\exp(-T_{\mathrm{d}}/T_1)$ on the same curve; if both branches continue and hand on their registers with their molecule-number weights (the closed carrier of Sec. S4), the curve itself changes (Table S18): the two branches carry polarisations of opposite sign and what survives their sum is the hyperfine evolution of the nuclear state, not the spin sorting.

TABLE S18. Memory against proton depolarisation q_{H} for the proton-only register on the closed carrier (^{14}N wiped every cycle, both branches handed on; six seeds).

q_{H}	MC (8 ch)	IPC (8 ch)
0.00	1.547 ± 0.078	2.560
0.10	1.459 ± 0.060	2.469
0.30	1.283 ± 0.040	2.288
0.50	1.148 ± 0.027	2.151
0.70	1.048 ± 0.021	2.050
0.80	1.017 ± 0.016	2.020
0.90	1.005 ± 0.007	2.007
1.00	1.003 ± 0.004	2.005

On the product-selected closure the Trp Hβ ceiling is 1.02 ms on the intramolecular term and 0.66 ms with the bath, the boundaries 1.55 and 1.01 ns, a required speed-up of 10× and 15×; the methyl's ceiling is 2.1 and 1.2 ms with boundaries 3.2 and 1.8 ns. On the closed carrier the register the biology keeps does not reach the band at any τ_c with the 0.5 memory threshold used throughout (the ceiling is 0.06 ms), and with the threshold relaxed to 0.2 the boundary sits at 0.40 and 0.26 ns, some 38–58 times faster than the protein. The conditional product register of Table S17 is therefore an upper bound; the number the criterion should be read with depends on which branch the chemistry regenerates, and the paper quotes both.

S10. ENSEMBLE AVERAGING

Every capacity above is a single-molecule quantity, whereas a downstream reaction reads a pool of order 10^{10} copies. A density matrix already averages over a *homogeneous* ensemble sharing a common clock, so that case is covered; what is not covered is heterogeneity between copies and desynchronisation of their turnover schedules. We compute both by pooling the product signal of many copies driven by a common input, and—as with the accumulated-pool route—we measure each pooled readout against its *own* floor, obtained by wiping the nuclear register every cycle. Only the excess over that floor is register memory.

Desynchronisation is the binding constraint. Copies are placed at phase offsets drawn uniformly in $[-\Delta, +\Delta]$ cycles (64 copies, three seeds). A copy offset by more than one cycle is showing an earlier *input*, so pooling itself becomes a delay line; the floor control is essential.

TABLE S19. Pooled capacity against turnover jitter Δ, as excess over the register-wiped floor of the same pooled readout.

Δ (cycles)	kinetics IPC	excess	CIDNP IPC	excess
0	1.960	+0.951	4.460	+2.463
0.1	1.698	+0.688	3.137	+1.528
0.25	1.526	+0.516	3.058	+1.437
0.5	1.009	+0.000	2.927	+1.306
1	1.117	+0.108	2.947	+1.326
2	1.829	+0.007	3.485	+1.019

The two readouts behave completely differently. The time-resolved kinetics lose almost all of their memory by half a cycle of jitter with the survivor register (+0.951 → +0.000) and reach zero near one and a half cycles with the product register (+1.746 → +0.353 at half a cycle, +0.250 at one, +0.009 at one and a half; Sec. S4). Those channels sample the cycle at particular times, so a pool spread over the cycle averages them into a single number. The survivor sequence is not monotonic—it returns to +0.110 at one cycle—and the $q = 1$ floor rises beyond one cycle, so these numbers should be read as the excess they are and not as a smooth decay curve. The apparent recovery of the raw IPC at $\Delta \geq 1$ is an artifact of pooling and not memory at all—at $\Delta = 2$ the excess over the register-wiped floor is +0.004. The CIDNP route, by contrast, retains half its excess even at $\Delta = 2$, because the product-weighted polarisation is integrated over the whole cycle and does not depend on when the pool is inspected. **Synchronised turnover is therefore a requirement of the kinetic readout specifically, not of the reservoir.** Of the two routes evaluated here, CIDNP is the one that survives a desynchronised ensemble; the remaining five routes of Table S4 were not evaluated under jitter, so we do not claim it is the only survivor among all seven.

Heterogeneity is not a threat. Scattering the three hyperfine couplings log-normally about the cryptochrome point does not degrade the pooled capacity; see `open1_ensemble_heterogeneity.json`. Mild spread slightly *improves* it, which we attribute to the averaging reducing the ill-conditioning of the nested time-resolved channels discussed above rather than to any spin-dynamical effect.

Heterogeneous relaxation. The heterogeneity above varies the Hamiltonian at a common relaxation rate, which is not the variable the register criterion turns on. A real pool does not tumble at one rate: some copies stay bound to the protein and some diffuse freely, so τ_c—and through it T_{1n}, and through that the fraction of the register surviving a turnover—differs from copy to copy. Because the readout sums over copies and cannot tell them apart, a fast-relaxing sub-population could in principle wash out a slow one's memory. We test this by drawing τ_c log-normally about a 5 ns median with log-spread σ, taking six stratified quantiles as the copies, and giving each its own survival $q_j = 1 - \exp(-T_\mathrm{d}/T_1(\tau_c^{(j)}))$ through the relaxation model of Sec. S12 at $T_\mathrm{d} = 10$ ms (three seeds, eight-channel CIDNP route).

TABLE S20. Pooling copies that relax at different rates. The mean-field column is a single homogeneous copy at the pool's own mean q; the last two columns are what the *spread* contributes once that is removed, in absolute terms and as a fraction of the rise over $\sigma = 0$.

σ	τ_c (ns)	$\bar{q}$	pooled excess	spread alone
0	5.00	0.739	+1.947	+0.000
0.3	3.30–7.57	0.735	+1.946	−0.003
0.6	2.18–11.46	0.722	+1.947	−0.008
1	1.25–19.93	0.696	+1.960	−0.003

The pooled excess does not move: +1.947 at $\sigma = 0$ and +1.946, +1.947 and +1.960 at $\sigma = 0.3$, 0.6 and 1—a change of 0.7 per cent at a sixteen-fold spread in τ_c, within the seed scatter. The mean survival $\bar{q}$ drifts from 0.739 to 0.696, because $q = 1-e^{-T_\mathrm{d}/T_1}$ saturates at unity: a copy displaced to slow τ_c can lose much more than the fast copy opposite it can gain, so a symmetric spread in τ_c produces an asymmetric, downward response in q (at $\sigma = 1$ the six values are 0.286 to 0.995, i.e. −0.45 and +0.26 about the median). The mean-field control—a uniform pool at the heterogeneous pool's own mean survival—reproduces the pooled excess to within 0.008 at every spread (residuals −0.003, −0.008, −0.003; the control reproduces the $\sigma = 0$ value to 10^{-12}, which is the consistency check that the decomposition is doing what it claims).

What this licenses is worth stating carefully. It licenses applying the criterion to a heterogeneous pool, and it licenses the cleaner statement too: to the precision of this scan a heterogeneous pool *is* a homogeneous pool at its mean survival, so a single effective T_{1n} is not merely an adequate working approximation but the quantity the pooled readout actually measures. The scan covers $\sigma \leq 1$ about a single median and does not speak to a pool split into two widely separated populations. It does not rescue the kinetic routes from desynchronisation, which is a separate axis and remains the binding ensemble constraint (`open1_ensemble_desync_floor.json`); the scan of this section is `open6_tau_c_heterogeneity.json`.

S11. ANISOTROPIC HYPERFINE COUPLING AND THE NUCLEAR SPIN QUANTUM NUMBER

The main text uses isotropic couplings and models ^{14}N as spin-1/2. We relax each in turn. Both scans use the same cycle, readout and estimator, and each is scored against its own $q = 1$ floor (three seeds).

Anisotropy. We hold the isotropic average fixed and grow an axial anisotropy, $\mathbf{A} = A_\mathrm{iso}(1-\eta, 1-\eta, 1+2\eta)$, at fixed molecular orientation. This is a *scan*, not a calculation with the published tensor sets [Lee *et al.*, J. R. Soc. Interface **11**, 20131063 (2014); Schleicher *et al.*, Sci. Rep. **11**, 18395 (2021)], and it bounds the sensitivity without removing the limitation.

TABLE S21. Capacity against axial anisotropy η (eight-channel CIDNP route).

η	IPC	floor	excess
0 (isotropic)	4.460	1.996	**2.463**
0.5	3.401	2.001	1.400
1.0	1.009	1.009	0.000
2.0	2.572	1.999	0.573

This is the least comfortable result in the paper. Realistic axial anisotropy costs two-fifths of the capacity (two-thirds with the product register, Sec. S4), and at $\eta = 1$ it vanishes exactly. The mechanism is transparent and is the same one that makes CIDNP the write channel: $\eta = 1$ is the pure-Ising point $A_{xx} = A_{yy} = 0$, where the hyperfine term commutes with I_z and S_z, so there are no flip-flops, no singlet–triplet mixing driven by the nuclei, and no polarisation transfer. The capacity is carried by the *perpendicular* components of the hyperfine tensor, and $\eta = 2$—where $|A_\perp|$ is restored with opposite sign—recovers to 0.573. A real flavin or tryptophan coupling has substantial $A_\perp$, so the physical conclusion is that the reservoir survives anisotropy with a substantially reduced capacity, and fails only in an Ising limit that no real radical realises. We report the reduction rather than quoting the isotropic number as though it were general.

^{14}N as spin-1. A four-spin reduction (e_1, e_2, one N, one H) is run with the nitrogen as spin-1/2 and then as spin-1, everything else fixed, so the difference isolates the spin quantum number.

TABLE S22. The spin-1/2 approximation for ^{14}N is conservative in the four-spin reduction (one N, one H); the five-spin model has not been run with spin-1 nitrogen.

model	dim	IPC	floor	excess
N as spin-1/2 (main text)	16	3.610	1.984	1.626
N as spin-1 (real ^{14}N)	24	3.682	1.997	**1.684**

Treating ^{14}N correctly *increases* the excess capacity

by 28%: the larger nuclear Hilbert space is a larger register. The approximation used in the main text therefore understates the reservoir, and no conclusion depends on it.

The published tensors. The scan above varies an axial anisotropy parameter at fixed isotropic average. The published tensor set for the cryptochrome pair—the DFT tensors (UB3LYP/EPR-III) of $FAD^{\bullet-}$ and $TrpH^{\bullet+}$ used by Lee *et al.* [J. R. Soc. Interface **11**, 20131063 (2014)] and tabulated in full, in one common axis system with the FAD/Trp-342 orientation of *Dm*Cry (PDB 4GU5), by Hiscock *et al.* [Proc. Natl. Acad. Sci. U.S.A. **113**, 4634 (2016), SI Appendix Tables S1 and S2]—is used in Table S23: first on the three nuclei of the main model (N5, N10, Trp Hβ1; isotropic parts 0.523, 0.189 and 1.605 mT, the last replacing the 40 MHz of the main text), with the field direction scanned in that frame and a six-direction average formed by pooling the channels of the six orientations as one ensemble readout, and then with a fourth nucleus, Trp H1 or FAD H6, on a three-axis average (six spins; $n_t = 48$, $L = 400$, three seeds, which costs about one per cent on MC, Sec. S7). Spin-1/2 nitrogen throughout; the product polarisation is read along the field and the electronic dephasing axis follows it, so the isotropic model is exactly orientation-invariant. The proton-only register wipes the nitrogens every cycle. Table S24 keeps the limiting case $A_\perp = 0$ (the idealisation Lee *et al.* identify as the one [$FAD^{\bullet-}$ $Z^{\bullet}$] nearly satisfies) and an in-plane component of a tenth to three tenths of a_{iso}, trace preserved.

TABLE S23. The cryptochrome point with the published hyperfine tensors. Excess over each case's own floor; kinetic (5 ch) and CIDNP routes, full and proton-only register (columns: route channels, register; the three-nucleus rows are N5, N10, Trp Hβ1; the six-spin rows add the named nucleus). Isotropic model of the main text: full register +1.75 and +2.96, proton-only +0.00 and +0.86.

nuclei	field	5, full	8, full	5, proton	8, proton
3 nuclei	$\theta = 0$	+0.897	+0.942	+0.896	+1.362
3 nuclei	$\theta = 54.7°$	+0.580	+1.320	+0.573	+1.032
3 nuclei	$\theta = 90°$	+0.087	+0.830	+0.042	+0.821
3 nuclei	6-dir.	+0.769	+0.935	+0.743	+0.603
+Trp H1	3-axis	+0.609	+0.774	+0.605	+0.984
+FAD H6	3-axis	+0.609	+0.983	+0.446	+1.474

Three things change under the published tensors. The full register's CIDNP excess falls from 2.96 to 0.93 in the orientation average: the near-axial nitrogens are written far less than isotropic ones of the same a_{iso}, as the limiting case below shows in its sharpest form. The proton-only register, the one the criterion rests on, retains seven tenths of its excess in the orientation average (0.60 against 0.86 isotropic, and 0.82–1.36 at the single orientations; 0.98 and 1.47 with a second proton). And the time-resolved kinetic route, which on the isotropic model returns exactly its floor once the nitrogens are wiped, retains 0.74 on the proton register here: with the nitrogens barely coupled to the mixing, the yield's time course is shaped by the tryptophan proton alone, and that proton persists. The register restriction of Sec. S9 is therefore a statement about the isotropic model where it says the kinetic route dies; under the published tensors the kinetic route survives on the proton register at a reduced level. The criterion's numbers, which rest on the proton register, are not loosened by any row of this table; the orientation average would tighten the register-reuse curve by about three tenths, within the factor the closed cycle already imposes.

TABLE S24. The cryptochrome point with the flavin nitrogens as axial tensors. Excess over each case's own floor, for the full and the proton-only register.

case	5 ch, full	8 ch, full	8 ch, proton
isotropic	+1.752	+2.957	+0.861
axial, $\theta = 0$	−0.000	+0.246	+0.591
axial, $\theta = 54.7°$	−0.003	−0.303	+0.754
axial, $\theta = 90°$	−0.000	+0.719	+0.706
axial, 6-dir. average	−0.003	−0.194	+0.749
$A_\perp/a_{\mathrm{iso}} = -0.1$, 6-dir. avg.	+0.004	−0.156	+0.738
$A_\perp/a_{\mathrm{iso}} = -0.2$, 6-dir. avg.	+0.091	−0.205	+0.836
$A_\perp/a_{\mathrm{iso}} = -0.3$, 6-dir. avg.	+0.518	+0.853	+0.920

With the nitrogens axial the full register of the main text is not representative: the kinetic route loses its memory at every orientation, and the eight-channel excess is +0.25 at $\theta = 0$ and negative (the channels fall below their own floor) at 54.7° and in the orientation average, recovering only when the in-plane component reaches three tenths of a_{iso}. An Ising-like nitrogen cannot be written by singlet–triplet mixing, which is the anisotropy result of the scan above in its sharpest form. The proton-only register, the one the criterion rests on, is insensitive to the nitrogen tensors: its excess is 0.75 in the orientation average and 0.74–0.92 across the in-plane scan, against 0.86 for the isotropic model. The criterion's numbers therefore do not depend on the nitrogen tensor form; the headline capacities of the full register do, and should be read as the isotropic model's.

S12. PREDICTING THE TWO UNMEASURED CONSTANTS

The criterion is stated in terms of $T_{\mathrm{turnover}}/T_{1n}$ and neither quantity has been measured for a neuronal cryptochrome. Neither can be measured here, but both can be *predicted* from measured constants, and doing so changes the conclusion. We validate the relaxation calculator against three independent measured systems before applying it to the unmeasured one.

TABLE S25. Validation of the relaxation calculator.

system	predicted	measured	ratio
^{2}H in D_2O (quadrupolar)	0.432 s	0.40 s	1.08
^{14}N of free amino acid	2.3 ms	2.9–8.8 ms	0.5
$^{1}H_2O$ intramolecular (dipolar)	5.55 s	3.6 s[1]	1.54

The relaxation time is a low-field quantity. For a 60 kDa protein at 310 K, Stokes–Einstein–Debye gives $\tau_c = 15.2$ ns. At the geomagnetic field $\omega\tau_c \approx 2\times10^{-4}$, so every nucleus here sits deep in extreme narrowing, where $1/T_1 \propto \tau_c$ is *maximal*. This is the point the $T_{1n} \approx 1$ s assumption misses: that value is a high-field number.

TABLE S26. Predicted T_1 of candidate register nuclei.

nucleus	at 50 μT	at 9.4 T
Trp Hβ (CH_2)	2.4 ms	9.0 s
flavin 8α-CH_3	5.0 ms	2.3 s
^{14}N5 / N10 (flavin)	0.34 / 0.19 μs	7 / 4 μs

The protein-bound proton relaxes in milliseconds at the field the organism actually experiences, not in seconds. Because $1/T_1 \propto \tau_c$ in this regime, the register's lifetime is governed by how fast the product reorients: T_1 rises to 37 ms at $\tau_c = 1$ ns and to 0.37 s at $\tau_c = 0.1$ ns.

The drive cannot be light. With the measured flavin extinction coefficient ($\varepsilon_{450} \approx 11\,300$ $M^{-1}cm^{-1}$, cross-section 4.3×10^{-17} cm^2) and a quantum yield of 0.1, a single flavin turns over once per 7.7 s in full sunlight, once per 77 s under overcast sky, and—for a blue-light attenuation of 10^{-9} through a centimetre of tissue—once per ~ 250 years inside the skull. A light-driven cryptochrome reservoir in a neuron fails by roughly ten orders of magnitude. The drive must therefore be catalytic, and a flavoenzyme turnover number of 1–1000 s^{-1} puts T_{turnover} at 1 ms–1 s, which is the right range.

Putting the two together. Requiring simultaneously that (a) the register be reused, (b) T_1 follow from the calculation above, and (c) the horizon $(\mathrm{MC} - C_0)T_{\mathrm{turnover}}$ land in the 10 ms–1 s band, and interpolating $\mathrm{MC}(q)$ from the measured reuse curve of Table S12:

TABLE S27. Feasible region, on the intramolecular term alone (upper block) and with the intermolecular proton bath (lower block). The retained memory is required to exceed 0.5, about four times the seed dispersion, so that the residual $\mathrm{MC}(q{=}1) - C_0 = 0.002$ cannot manufacture a horizon.

τ_c	T_1 (50 μT)	viable T_{turn}	horizon
intramolecular term alone			
0.1 ns	372 ms	5.6–1.22×10^{3} ms	827 ms
1 ns	37 ms	6.6–122 ms	83 ms
5 ns	7.4 ms	10–24.4 ms	17 ms
10 ns	3.7 ms	none	8.3 ms
15.2 ns (bound)	2.4 ms	**none**	**5.4 ms**
with the intermolecular bath, $f = 0.54$			
0.1 ns	241 ms	5.6–792 ms	536 ms
1 ns	24 ms	7.3–79.2 ms	54 ms
5 ns	4.8 ms	10–13.9 ms	11 ms
10 ns	2.4 ms	none	5.4 ms
15.2 ns (bound)	1.6 ms	**none**	**3.5 ms**

Two corrections are needed before this table can be read as a boundary, and they pull in opposite directions.

The boundary is solved, not read off the grid. The scan above samples τ_c at $\ldots, 5, 10, 15.2$ ns and never evaluates anything between 5 and 10, so quoting "feasible only for $\tau_c \le 5$ ns" reports the last sampled point rather than the boundary. Bisecting on the same feasibility predicate puts it at 8.3 ns, a requirement of 1.8× rather than 3×.

The intermolecular bath. A T_1 that counts only the intramolecular partner is an upper bound. The validation table above supplies the scale of that omission: liquid water's intramolecular-only prediction is 5.55 s against a measured total of 3.6 s, a bath contributing 54% of the intramolecular rate. We use that residual as a calibration, not as an independent measurement—f is defined from it, so the ratio it restores to unity is an identity. Its transplant onto a protein interior is defensible by magnitude rather than by mechanism: the resulting $\Sigma_{\mathrm{ext}} = 0.017$ Å^{-6} sits at the low end of the 0.016–0.030 a uniform protein proton density gives for closest approaches of 2.5–2.0 Å, which is the conservative end for a claim of exclusion. Including it shortens T_1 for the protein-bound proton from 2.4 to 1.6 ms, the ceiling from 5.4 to 3.5 ms, and moves the boundary from 8.3 to 5.4 ns—a requirement of 2.8× where the intramolecular term alone would ask 1.8×.

A register whose dipolar vector reorients only with the intact protein cannot reach the neural band: its ceiling is 5.4 ms on the intramolecular term alone and 3.5 ms with the bath, below the band under every turnover rate. The criterion is a statement about $\tau_{\mathrm{eff}} = S^2\tau_c + (1-S^2)\tau_{\mathrm{int}}$ and about the total dipolar sum, not about chemical binding. The flavin 8α-CH_3 makes the distinction concrete: its intramethyl H–H vector has the exact geometric value $S^2 = 1/4$ and therefore does

not co-rotate with the protein despite being covalently attached, which on the intramolecular term alone would give $T_1 = 5.0$ ms and a ceiling of 11.2 ms—inside the band. Its own boundary sits at 17.1 ns without the bath, i.e. it would be feasible even at the protein's own tumbling rate. With the bath that boundary falls to 9.6 ns and the escape closes. The two candidate nuclei do not land at the same requirement—the methyl's boundary is 9.6 ns against 5.4 ns for Trp Hβ, because its intramethyl vector is already partly averaged—but both fall below the protein's own 15.2 ns, the methyl by a factor of 1.6. On the proton-only product register (Sec. S9) both candidates tighten: Trp Hβ to 4.0 ns without the bath and 2.6 ns with it, the methyl to 8.2 and 4.6 ns, and the methyl's ceiling falls to 5.4 and 3.0 ms, below the band either way.

How much of this depends on the band edge. The requirement scales linearly with where the neural band is taken to start. In extreme narrowing $1/T_1 \propto \tau_{\mathrm{eff}}$, so the ceiling is exactly inversely proportional to τ_{eff} and the required speed-up is

$$\text{speed-up} = 1.84\,(1+f) \times \frac{\text{band edge}}{10\ \text{ms}},$$

with f the intermolecular bath ratio. Bisecting numerically agrees with the closed form to two decimals. A register that reorients only with the protein becomes feasible once the band edge falls below 5.4 ms on the intramolecular term alone, but below only 3.5 ms once the bath is included. Taking the band to start at 5 ms rather than 10 ms therefore does *not* rescue it. We state this because the 10 ms edge is a convention, and the conclusion should be read as conditional on it.

TABLE S28. Robustness of the ceiling for the Trp Hβ vector ($S^2 = 1$). Each listed variation leaves the horizon below the 10 ms band edge. The table varies the tumbling time, the partner count and the memory threshold; it does not vary the bath ratio f or the bath order parameter, whose sensitivity is given in the text, nor S^2 itself, which is the separate methyl case above.

variation	τ_c (ns)	T_1 (ms)	ceiling (ms)
as published, hydration 1.3	15.24	2.44	5.43
hydration 1.0	11.73	3.17	7.05
hydration 1.6	18.76	1.98	4.41
viscosity ×2	30.49	1.22	2.71
viscosity ×4	60.97	0.61	1.36
120 kDa dimer	30.49	1.22	2.71
2 dipolar partners	15.24	1.22	2.71
3 dipolar partners	15.24	0.81	1.81
memory threshold 0.2	15.24	2.44	5.43
memory threshold 1.0	15.24	2.44	2.88
+ bath ($f = 0.54$)	15.24	1.58	3.52

Several of these shorten the ceiling further. The one that would *lengthen* it, and so weaken the exclusion, is a rank error: τ_{rot} here is the *rank-2* correlation time, $1/(6D_r)$, which is what the dipolar spectral densities require. The rank-1 value used in dielectric and fluorescence-anisotropy work is three times longer (45.7 ns), and feeding it into $J(\omega)$ would give $T_1 = 0.81$ ms and a ceiling of 1.8 ms. That mistake would strengthen the exclusion, so we name the rank in the function itself rather than rely on a reader catching it.

This converts the paper's weakest statement—a criterion whose two parameters were simply unknown—into a quantitative and falsifiable one:

> *The radical pair must be regenerated catalytically, not photochemically, on a register that reorients with $\tau_c \lesssim 2.6$ ns (5.4 ns if all three nuclei could be kept) while remaining part of the same molecule, at a turnover interval drawn from the corresponding row of Table S27 with the bath included:* 10–14 *ms at* $\tau_c = 5$ *ns,* 7.3–79 *ms at* 1 *ns,* 5.6–790 *ms at* 0.1 *ns. The two conditions are not independent, and quoting a single interval alongside a single* τ_c *bound would overstate the region.*

Both halves are measurable with existing methods: τ_c by NMR relaxation dispersion on the isolated protein, and the turnover number by steady-state kinetics. Neither requires a neuron.

S13. SPIN PARAMETERS

Each radical carries its own nuclei; they are not shared, which is what a spatially separated pair means. The Hamiltonian therefore contains three hyperfine terms, two on the first radical and one on the second.

TABLE S29. Hyperfine assignment at the cryptochrome point. Values are representative isotropic magnitudes of the dominant $\mathrm{FAD}^{\bullet-}$ and $\mathrm{TrpH}^{\bullet+}$ couplings; the full anisotropic tensors are given by Lee *et al.*, J. R. Soc. Interface **11**, 20131063 (2014), and the flavin-radical ENDOR couplings by Schleicher *et al.*, Sci. Rep. **11**, 18395 (2021).

coupling	radical / nucleus	(mT)	(MHz)
A_{e_1a}	$\mathrm{FAD}^{\bullet-}$, ^{14}N5	~ 0.523	14.66
A_{e_1b}	$\mathrm{FAD}^{\bullet-}$, ^{14}N10	~ 0.189	5.30
A_{e_2a}	$\mathrm{TrpH}^{\bullet+}$, Hβ1	~ 1.43	40

The engineered operating point uses $A_{e_1a} = 80$, $A_{e_1b} = 40$, $A_{e_2a} = 15$ MHz, $J = 2$ MHz, $B = 1$ mT and $\tau = T_2^e = 50$ ns. The conclusions depend on these magnitudes lying in the few–40 MHz range typical of flavin and tryptophan radicals, not on the precise values. The register criterion is *not* independent of the hyperfine assignment, however: Sec. S11 shows that the capacity is carried by the perpendicular components of the hyperfine tensor and vanishes when they do.

S14. DEPENDENCE OF THE CRYPTOCHROME POINT ON THE ELECTRON–ELECTRON COUPLING

The cryptochrome operating point of the main text takes $J = 0$ with no dipolar term: the two radicals interact only through recombination. The source cited for the separated pair (Efimova and Hore, *Biophys. J.* **94**, 1565 (2008)) reports, for the FAD–W324 pair of cryptochrome at a centre-to-centre distance $r = 1.90$ nm, a point-dipole coupling $D = -2.78 \times 10^3\,\mu\mathrm{T}/(r/\mathrm{nm})^3 = -11.36$ MHz (the $T_{\pm}$–T_0 splitting; the full-tensor value differs by under 5%) and an exchange coupling $J = J_0 e^{-\beta r}$ with $\beta = 14$ nm^{-1} and $8 \times 10^{12} \le |J_0|/\mu\mathrm{T} \le 8 \times 10^{14}$, i.e. an S–T gap of 1.3–126 MHz whose sign that work does not determine, the two partially cancelling near 2 nm. This section reruns every cryptochrome-point quantity the paper quotes with those values in the Hamiltonian and nothing else changed: same protocol, estimator, seeds and sequence lengths. D enters as the axial tensor $2D(\boldsymbol{S}_1\cdot\hat{n})(\boldsymbol{S}_2\cdot\hat{n}) - \frac{2D}{3}\boldsymbol{S}_1\cdot\boldsymbol{S}_2$ (so that at $B = 0$ without hyperfine coupling the triplet levels sit at $D/3$ and $-2D/3$, the Efimova–Hore convention), with the inter-radical axis $\hat{n}$ parallel and perpendicular to B; no powder average is taken. J_{gap} is the S–T gap entered as the J of $J\,\boldsymbol{S}_1\cdot\boldsymbol{S}_2$, and both signs are run because neither is preferred by the source. All numbers are from `qbscreen/premise_dependence.py` (six seeds, mean ± s.d.).

Capacity without recombination. Table S30 is the calculation behind the statement that recombination is what writes the input into the register: with recombination switched off, the reference premise gives the memoryless floor. It is the only point in the table that does. Any electron–electron coupling—0.6 MHz of D at $r = 5$ nm, or 1.3 MHz of J—restores a capacity of 5–6, because coupled electrons transfer the singlet–triplet correlation to the nuclei without recombining. The reference premise is therefore the one point of the coupling range at which recombination is the *only* write mechanism; everywhere else it is one of two.

TABLE S30. Cryptochrome-point capacity with recombination switched off (full observable set, $L = 900$, $T_2^e = 1$ μs), as a function of the electron–electron coupling. J and D in MHz; $\hat{n}$ is the inter-radical axis relative to B.

coupling	J	D	$\hat{n}$	IPC
reference	+0.00	0.00	∥	1.02 ± 0.01
D only	+0.00	−11.36	∥	5.68 ± 0.17
D only	+0.00	−11.36	⊥	5.88 ± 0.05
D and J_{gap}	−12.60	−11.36	∥	5.60 ± 0.18
D and J_{gap}	−12.60	−11.36	⊥	5.09 ± 0.10
D and J_{gap}	+12.60	−11.36	∥	5.52 ± 0.09
D and J_{gap}	+12.60	−11.36	⊥	5.74 ± 0.13
J only, low end	+1.26	0.00	∥	5.72 ± 0.16
J only, mid-range	+12.60	0.00	∥	5.73 ± 0.15
J only, high end	+126.00	0.00	∥	5.54 ± 0.23
D only, $r = 2.2$ nm	+0.00	−7.32	∥	5.48 ± 0.28
D only, $r = 3.5$ nm	+0.00	−1.82	∥	5.71 ± 0.10
D only, $r = 5.0$ nm	+0.00	−0.62	∥	5.24 ± 0.12

Readout routes. Table S31 repeats the route survey of Sec. S4 under each premise, with each route's memoryless floor recomputed at $q = 1$. The floors do not move. The end-point yield stays on its floor under every premise: it remembers nothing whatever the coupling. The time-resolved kinetic routes move in both directions, from 2.0 to 1.3–3.4, because the coupled electrons now write through the yield as well as through recombination, and orientation and the sign of J decide whether that helps or hinders. CIDNP spans 3.9–5.2 against 4.42, the perpendicular orientation lowering it and the parallel one raising it; its excess over its own floor is 1.9–3.2 against 2.42. With the register restricted to the proton (both ^{14}N depolarised every cycle, as in Sec. S9) the kinetic route falls to its floor under the coupled premise exactly as under the reference one (1.02), and CIDNP retains 3.29 (D, ∥) and 2.86 (D, $J_{\mathrm{gap}} = -12.6$, ∥) against 2.78 under the reference premise, so the conclusion that only CIDNP survives the proton restriction is premise-independent. The classical baselines of Sec. S6 are unchanged by construction, and the largest quantum-side value in the table, 3.39 at five channels and 5.20 at eight, remains below the matched network (4.47 and 6.94): there is no premise in the range at which a quantum advantage appears.

Clock and register reuse. Table S32 repeats the turnover-clock scan of Sec. S7 ($T_{1n} = 1$ s, six seeds) and the register-reuse scan under the coupled premise. The linear memory capacity of the CIDNP route at zero pause moves from 2.92 to 2.72 (D, ∥) and 2.88 (D, $J_{\mathrm{gap}} = -12.6$, ∥), and falls with the pause as before. The MC(q) curve is flatter near $q = 1$ under the coupled premise: at $q = 0.99$ it retains 1.78 against 1.04, because the coupled electrons write more polarisation per cycle, so the one per cent of the register that survives still carries a readable signal. Fed into the horizon ceiling $\max_{T_d}[\mathrm{MC}(q(T_d)) - 1]\,T_d$ with the relaxation times

TABLE S31. Route capacities (IPC, six seeds) under the reference premise and under the Efimova–Hore couplings at $r = 1.90$ nm, for both signs of the exchange gap and both orientations of the inter-radical axis. Seed standard deviations are 0.03–0.27 and are given in the shipped file.

route	reference	D, $\parallel$	D, $\perp$	D, $J_{\mathrm{gap}} = -12.6$, $\parallel$	D, $J_{\mathrm{gap}} = -12.6$, $\perp$	D, $J_{\mathrm{gap}} = +12.6$, $\parallel$	D, $J_{\mathrm{gap}} = +12.6$, $\perp$
end-point singlet yield (1 ch)	1.02	1.01	1.01	1.02	1.02	1.01	1.02
time-resolved kinetics (5 ch)	1.97	2.17	1.91	2.53	1.34	3.39	1.78
both exit channels, end point (2 ch)	1.28	1.65	1.23	1.66	1.02	1.77	1.09
both exit channels, time-resolved (10 ch)	1.97	2.19	2.20	2.58	1.35	3.42	1.84
three micro-environments (3 ch)	2.72	2.92	2.91	2.92	2.80	2.81	2.80
kinetics + CIDNP (8 ch)	4.42	4.31	4.02	4.71	3.90	5.20	4.16
floor, CIDNP route ($q = 1$)	2.00	2.00	2.01	1.97	1.98	2.01	2.01

of Sec. S12, and using the same eight-point scan and interpolation for both premises so that the comparison is like for like, the ceiling for a register that reorients with the intact protein moves from 5.2 to 8.8 ms on the intramolecular term alone and from 3.4 to 5.7 ms with the intermolecular bath—below the 10 ms band in every case—and the correlation time at which the band opens moves from 8.0 to 13.4 ns (intramolecular) and from 5.2 to 8.7 ns (with bath). The register must still reorient faster than the protein, by a factor of 1.8 with the bath against 2.9 under the reference premise. The main text quotes the refined twenty-one-point scan for the reference premise; the eight-point values above differ from it by the interpolation, not by the physics, and the coupled-premise scan was not refined.

TABLE S32. Turnover clock and register reuse under the coupled premise. Top: linear memory capacity of the CIDNP route against the pause T_d ($T_{1n} = 1$ s; the reference column is the twelve-seed scan of Table S11, the coupled columns six seeds). Bottom: MC (8 ch) and the kinetic IPC (5 ch) against the depolarisation q for the reference premise and for D, $\parallel$.

T_d	reference	D, $\parallel$	D, $J_{\mathrm{gap}} = -12.6$, $\parallel$
1 μs	2.92	2.72 ± 0.06	2.88 ± 0.07
1 ms	2.92	2.72 ± 0.06	2.88 ± 0.07
10 ms	2.89	2.71 ± 0.06	2.87 ± 0.07
100 ms	2.68	2.67 ± 0.06	2.83 ± 0.05

q	MC (ref.)	MC (D, $\parallel$)	IPC_5 (ref.)	IPC_5 (D, $\parallel$)
0.00	2.85	2.72	1.97	2.17
0.10	2.62	2.67	1.96	2.13
0.30	2.33	2.55	1.95	2.07
0.50	2.09	2.31	1.91	2.01
0.70	2.00	2.05	1.72	1.88
0.90	1.93	1.98	1.17	1.28
0.99	1.04	1.78	1.01	1.02
1.00	1.00	1.00	1.01	1.02